

Internet of Things – Project 4

Edge-Computing Smart Home Appliance (Interrupts & Safety)

Objective

To implement an ESP32-based smart home safety system using a PIR motion sensor, gas sensor, LEDs, buzzer, and LCD with interrupt-driven event handling and fail-safe protection.

Implementation

- Configured PIR interrupt.
- Read gas sensor continuously.
- Controlled LEDs and buzzer.
- Displayed status on LCD.
- Activated emergency override during gas leakage.

Source Code

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

#define PIR_PIN    27
#define GAS_PIN    34

#define GREEN_LED   25
#define RED_LED     26
#define BUZZER_PIN  33

#define GAS_THRESHOLD 2500

LiquidCrystal_I2C lcd(0x27, 16, 2);

volatile bool motionDetected = false;

void IRAM_ATTR motionISR() {
    motionDetected = true;
}

void setup() {
```

```

Serial.begin(115200);

pinMode(PIR_PIN, INPUT);
pinMode(GREEN_LED, OUTPUT);
pinMode(RED_LED, OUTPUT);
pinMode(BUZZER_PIN, OUTPUT);

digitalWrite(GREEN_LED, LOW);
digitalWrite(RED_LED, LOW);
digitalWrite(BUZZER_PIN, LOW);

lcd.init();
lcd.backlight();

lcd.setCursor(0,0);
lcd.print("Smart Home");
lcd.setCursor(0,1);
lcd.print("Initializing");
delay(2000);
lcd.clear();

attachInterrupt(
  digitalPinToInterrupt(PIR_PIN),
  motionISR,
  RISING
);

Serial.println("Smart Home Safety System Started");
}

void loop() {

  int gasValue = analogRead(GAS_PIN);

  Serial.print("Gas Sensor Value : ");
  Serial.println(gasValue);

  if (gasValue < 1000) {

    Serial.println("Status      : Air Quality Excellent");
  }
}

```

```

    Serial.println("Gas Level      : Very Low");
    Serial.println("Environment    : Safe");

}
else if (gasValue >= 1000 && gasValue < 1800) {

    Serial.println("Status        : Air Quality Normal");
    Serial.println("Gas Level      : Low");
    Serial.println("Environment    : Safe");

}
else if (gasValue >= 1800 && gasValue < GAS_THRESHOLD) {

    Serial.println("Status          : Gas Concentration Rising");
    Serial.println("Gas Level      : Moderate");
    Serial.println("Environment    : Monitor Carefully");

}
else if (gasValue == GAS_THRESHOLD) {

    Serial.println("Status          : Gas at Threshold");
    Serial.println("Gas Level      : Warning");
    Serial.println("Environment    : Take Precaution");

}
else {

    Serial.println("Status          : HIGH GAS DETECTED");
    Serial.println("Gas Level      : Dangerous");
    Serial.println("Environment    : EVACUATE IMMEDIATELY");

}

if (gasValue > GAS_THRESHOLD) {

    digitalWrite(GREEN_LED, LOW);
    digitalWrite(RED_LED, HIGH);
    digitalWrite(BUZZER_PIN, HIGH);

    lcd.clear();

```

```
lcd.setCursor(0,0);
lcd.print("!!! DANGER !!!");
lcd.setCursor(0,1);
lcd.print("GAS DETECTED");

Serial.println("Gas Leak Detected!");

motionDetected = false;

}

else {

    digitalWrite(RED_LED, LOW);
    digitalWrite(BUZZER_PIN, LOW);

    if (motionDetected) {

        digitalWrite(GREEN_LED, HIGH);

        lcd.clear();
        lcd.setCursor(0,0);
        lcd.print("Motion Found");
        lcd.setCursor(0,1);
        lcd.print("Lights ON");

        Serial.println("Motion Detected");

        delay(5000);

        digitalWrite(GREEN_LED, LOW);

        lcd.clear();
        lcd.setCursor(0,0);
        lcd.print("No Motion");
        lcd.setCursor(0,1);
        lcd.print("Lights OFF");

        motionDetected = false;
    }
}
```

```

else {

    lcd.setCursor(0,0);
    lcd.print("No Motion  ");
    lcd.setCursor(0,1);
    lcd.print("Lights OFF  ");

}

}

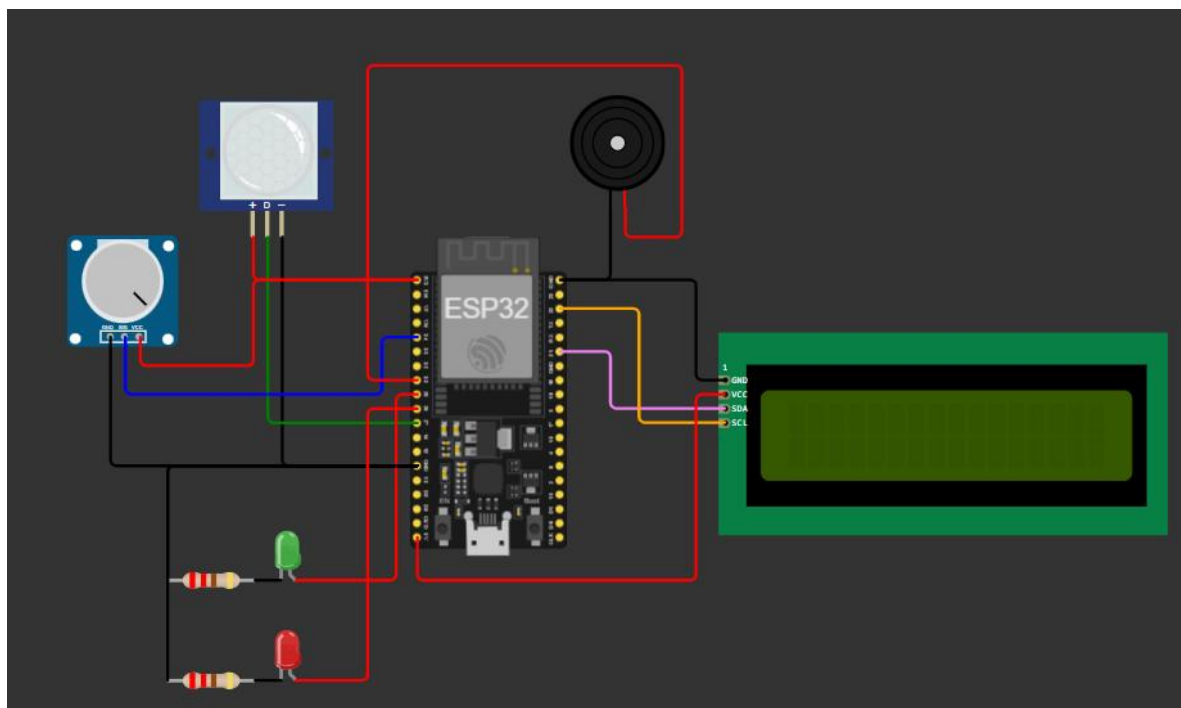
delay(200);

}

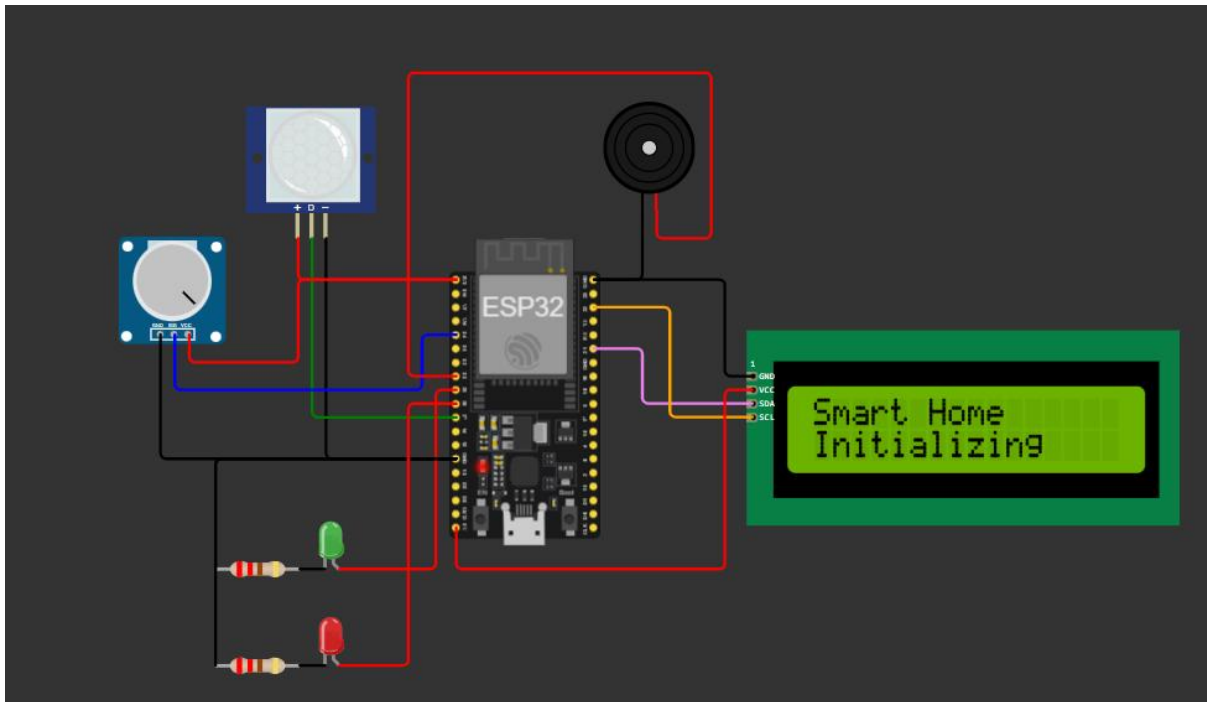
```

Result

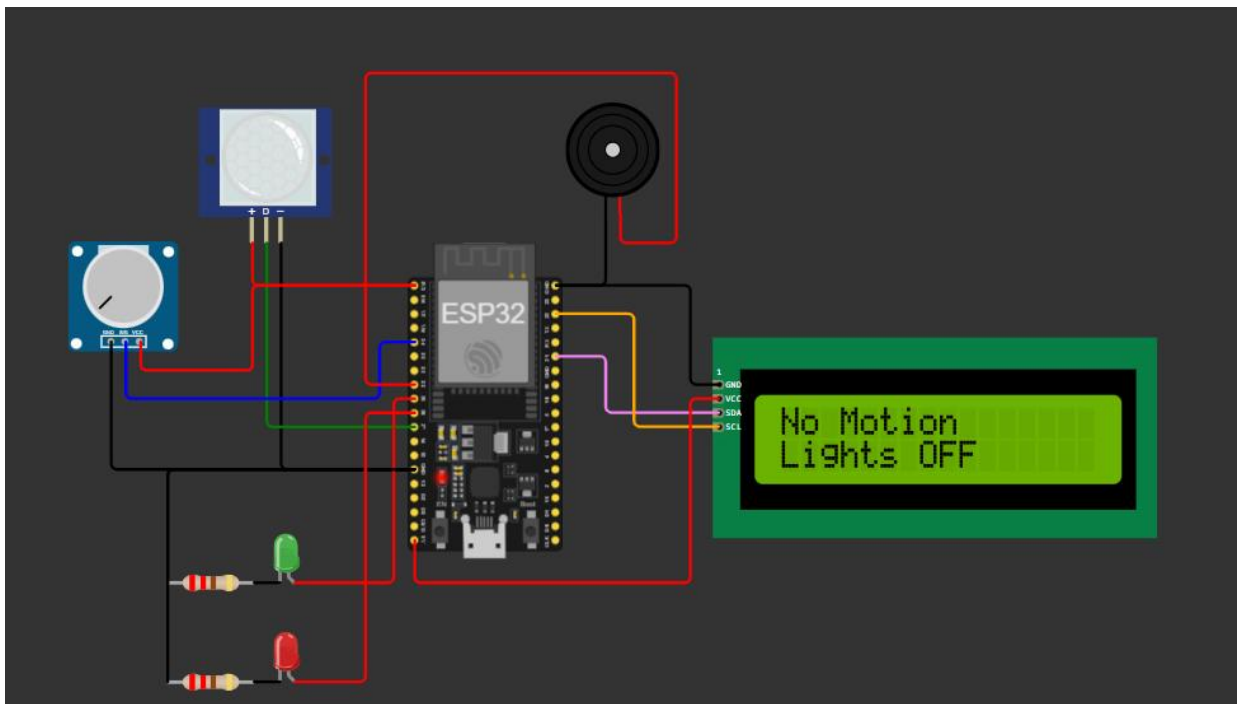
Hardware Setup



System Output

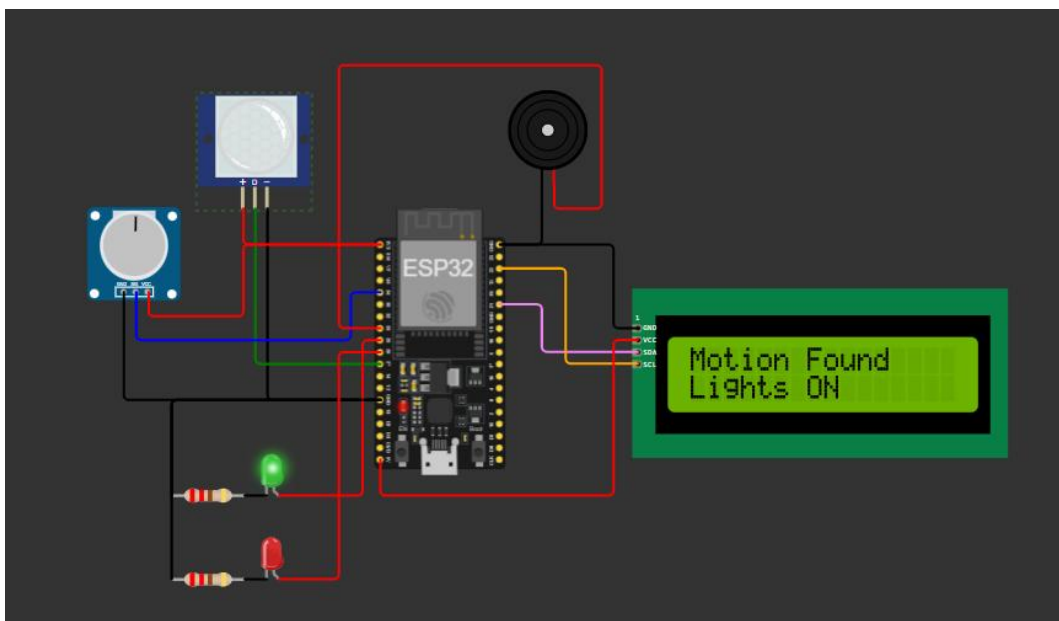


Hardware connections were completed successfully.



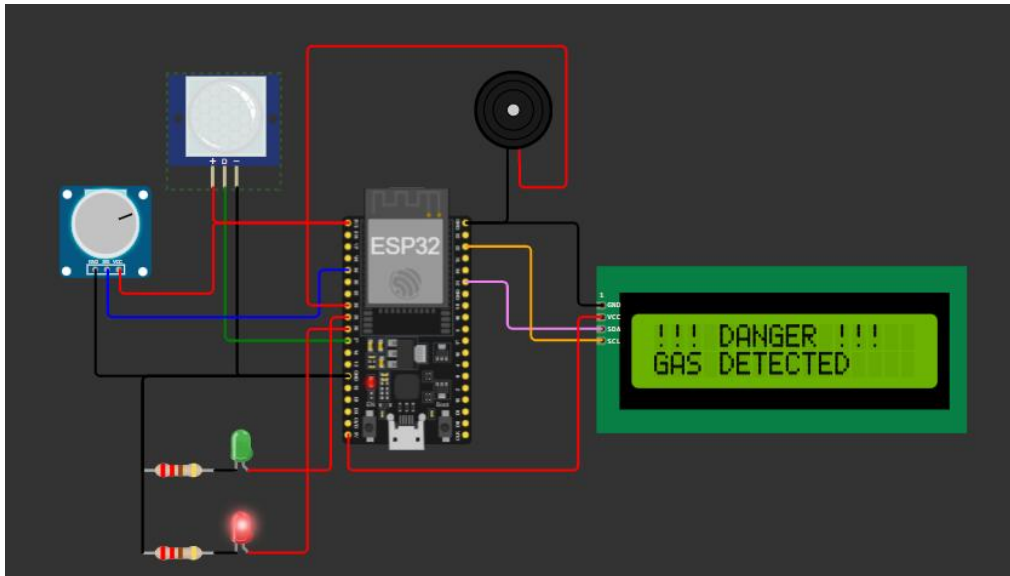
```
Status      : Air Quality Excellent
Gas Level   : Very Low
Environment : Safe
Gas Sensor Value : 0
Status      : Air Quality Excellent
Gas Level   : Very Low
Environment : Safe
Gas Sensor Value : 0
```

There is no Motion, there is no Detection



```
Gas Sensor Value : 2062
Status           : Gas Concentration Rising
Gas Level       : Moderate
Environment      : Monitor Carefully
Motion Detected
Gas Sensor Value : 2062
Status           : Gas Concentration Rising
Gas Level       : Moderate
Environment      : Monitor Carefully
```

Motion Detected, Green Light are turned ON



```
Status      : HIGH GAS DETECTED
Gas Level    : Dangerous
Environment  : EVACUATE IMMEDIATELY
Gas Leak Detected!
Gas Sensor Value : 2262
Status      : Gas Concentration Rising
Gas Level    : Moderate
Environment  : Monitor Carefully
Gas Sensor Value : 4095
Status      : HIGH GAS DETECTED
Gas Level    : Dangerous
Environment  : EVACUATE IMMEDIATELY
Gas Leak Detected!
```

Gas Leak Detected, Red light is turned ON

Conclusion

The smart home safety system successfully combined hardware interrupts with gas leak detection to provide reliable real-time monitoring and immediate emergency response. The project demonstrated the practical implementation of fail-safe edge-computing concepts in embedded IoT systems.